

ALFAA97045

## Cobalt(II,III) oxide, 99.5% (metals basis)

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | Cobalt(II,III) oxide, 99.5% (metals basis)<br>Cobalt(II,III) oxide, 99.5% (metals basis)                                                                                                                                                                           |
| Cat No. :                     | 97045                                                                                                                                                                                                                                                              |
| Synonyms                      | Tricobalt tetraoxide; Cobaltic-cobaltous oxide; Cobaltous oxide                                                                                                                                                                                                    |
| CAS No                        | 1308-06-1                                                                                                                                                                                                                                                          |
| Molecular Formula             | O4Co3                                                                                                                                                                                                                                                              |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                            |                         |                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------|
| Physical State<br>Powder Solid                                                                                                                                                             | Appearance<br>Dark grey | Odor<br>No information available |
| <b>Emergency Overview</b><br>May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause cancer by inhalation. Harmful to aquatic life with long lasting effects. |                         |                                  |

#### Classification of the substance or mixture

|                           |             |
|---------------------------|-------------|
| Respiratory Sensitization | Category 1  |
| Carcinogenicity           | Category 1B |
| Chronic aquatic toxicity  | Category 3  |

#### Label Elements



Signal Word

Danger

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

**Hazard Statements**

H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled

H350i - May cause cancer by inhalation

H412 - Harmful to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - In case of inadequate ventilation wear respiratory protection

**Response**

P304 + P341 - IF INHALED: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause cancer by inhalation.

**Environmental hazards**

Harmful to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component            | CAS No    | Weight % |
|----------------------|-----------|----------|
| Cobalt oxide (Co3O4) | 1308-06-1 | 100      |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component                                      | ACGIH TLV                   | OSHA PEL | NIOSH | The United Kingdom                 | European Union |
|------------------------------------------------|-----------------------------|----------|-------|------------------------------------|----------------|
| Cobalt oxide (Co <sub>3</sub> O <sub>4</sub> ) | TWA: 0.02 mg/m <sup>3</sup> |          |       | STEL: 0.3 mg/m <sup>3</sup> 15 min |                |

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

|  |  |  |  |                                                |  |
|--|--|--|--|------------------------------------------------|--|
|  |  |  |  | TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. |  |
|--|--|--|--|------------------------------------------------|--|

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

### Exposure Controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Dark grey                |                                          |
| <b>Physical State</b>                          | Powder Solid             |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      |                          |                                          |
| <b>Melting Point/Range</b>                     | 895 °C / 1643 °F         |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flash Point</b>                             | Not applicable           | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No information available |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>              | 6.11                     |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Water Solubility</b>                        | Insoluble                |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Autoignition Temperature</b>                | Not applicable           |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |
| <b>Molecular Formula</b>                       | O4Co3                    |                                          |
| <b>Molecular Weight</b>                        | 240.80                   |                                          |

### SECTION 10. STABILITY AND REACTIVITY

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>Stability</b>                | Stable under normal conditions. |
| <b>Hazardous Reactions</b>      | None under normal processing.   |
| <b>Hazardous Polymerization</b> | No information available.       |
| <b>Conditions to Avoid</b>      | None known.                     |
| <b>Materials to avoid</b>       | No information available.       |

**Hazardous Decomposition Products** None under normal use conditions.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

##### (a) acute toxicity;

| Component            | LD50 Oral             | LD50 Dermal               | LC50 Inhalation              |
|----------------------|-----------------------|---------------------------|------------------------------|
| Cobalt oxide (Co3O4) | LD50 > 5 g/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | LC50 > 5.06 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

| <b>Respiratory</b>                                                                                                                                                            | Category 1                                                                                                                                                                            |    |         |         |      |                      |  |  |        |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|---------|------|----------------------|--|--|--------|--|--|
| <b>Skin</b>                                                                                                                                                                   | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
|                                                                                                                                                                               | No information available                                                                                                                                                              |    |         |         |      |                      |  |  |        |  |  |
| <b>(e) germ cell mutagenicity;</b>                                                                                                                                            | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
| <b>(f) carcinogenicity;</b>                                                                                                                                                   | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
|                                                                                                                                                                               | The table below indicates whether each agency has listed any ingredient as a carcinogen                                                                                               |    |         |         |      |                      |  |  |        |  |  |
| <table><tr><th>Component</th><th>EU</th><th>UK</th><th>Germany</th><th>IARC</th></tr><tr><td>Cobalt oxide (Co3O4)</td><td></td><td></td><td>Cat. 2</td><td></td></tr></table> | Component                                                                                                                                                                             | EU | UK      | Germany | IARC | Cobalt oxide (Co3O4) |  |  | Cat. 2 |  |  |
| Component                                                                                                                                                                     | EU                                                                                                                                                                                    | UK | Germany | IARC    |      |                      |  |  |        |  |  |
| Cobalt oxide (Co3O4)                                                                                                                                                          |                                                                                                                                                                                       |    | Cat. 2  |         |      |                      |  |  |        |  |  |
| <b>(g) reproductive toxicity;</b>                                                                                                                                             | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
| <b>(h) STOT-single exposure;</b>                                                                                                                                              | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
| <b>(i) STOT-repeated exposure;</b>                                                                                                                                            | No data available                                                                                                                                                                     |    |         |         |      |                      |  |  |        |  |  |
| <b>Target Organs</b>                                                                                                                                                          | No information available.                                                                                                                                                             |    |         |         |      |                      |  |  |        |  |  |
| <b>(j) aspiration hazard;</b>                                                                                                                                                 | Not applicable                                                                                                                                                                        |    |         |         |      |                      |  |  |        |  |  |
|                                                                                                                                                                               | Solid                                                                                                                                                                                 |    |         |         |      |                      |  |  |        |  |  |
| <b>Symptoms / effects,both acute and delayed</b>                                                                                                                              | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |    |         |         |      |                      |  |  |        |  |  |

## SECTION 12. ECOLOGICAL INFORMATION

|                            |                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b> | The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Contains a substance which is: Harmful to aquatic organisms. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component            | Freshwater Fish                                  | Water Flea                            | Freshwater Algae                                       | Microtox |
|----------------------|--------------------------------------------------|---------------------------------------|--------------------------------------------------------|----------|
| Cobalt oxide (Co3O4) | LC50: > 136 mg/L, 96h static (Brachydanio rerio) | EC50: > 136 mg/L, 48h (Daphnia magna) | EC50: = 88 mg/L, 72h (Pseudokirchneriella subcapitata) |          |

|                                              |                                                                                                                 |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b>         |                                                                                                                 |
| <b>Persistence</b>                           | Insoluble in water.                                                                                             |
| <b>Degradability</b>                         | Not relevant for inorganic substances.                                                                          |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |
| <b>Bioaccumulative Potential</b>             | May have some potential to bioaccumulate                                                                        |
| <b>Mobility in soil</b>                      | Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility        |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors                                       |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance                                                  |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance                                                  |

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

### SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                           |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                |

### SECTION 14. TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Road and Rail Transport</b>      | Not Regulated                   |
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

### SECTION 15. REGULATORY INFORMATION

#### International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component            | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Cobalt oxide (Co3O4) | -                                                   | -                                       | X    | X     | 215-157-2 | X    | X   | X     | X    | X    | X    | KE-34112 |

#### National Regulations

### SECTION 16. OTHER INFORMATION

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
| <b>Revision Date</b>    | 13-May-2024                                        |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

#### Training Advice

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

# SAFETY DATA SHEET

## Cobalt(II,III) oxide, 99.5% (metals basis)

First aid for chemical exposure, including the use of eye wash and safety showers.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**